

⚡ Semantics Matter! Lessons From Stabilizing OpenTelemetry's System Conventions

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Observability Day 2026

Lesson 1

Naming is Hard

275 contributors

9 reviews/PR

mem_used_bytes

used_memory

system.memory.usage{state=used}

`os.type: unix, windows`

`os.type: linux, freebsd, darwin, ...`

Lesson 2

Abstractions Shape Your Metrics

process.pid

process.linux.cgroup

system.memory.usage

system.memory.linux.slab.usage

`system.cpu.usage`

`system.cpu.utilization`

`system.cpu.time`

Lesson 3

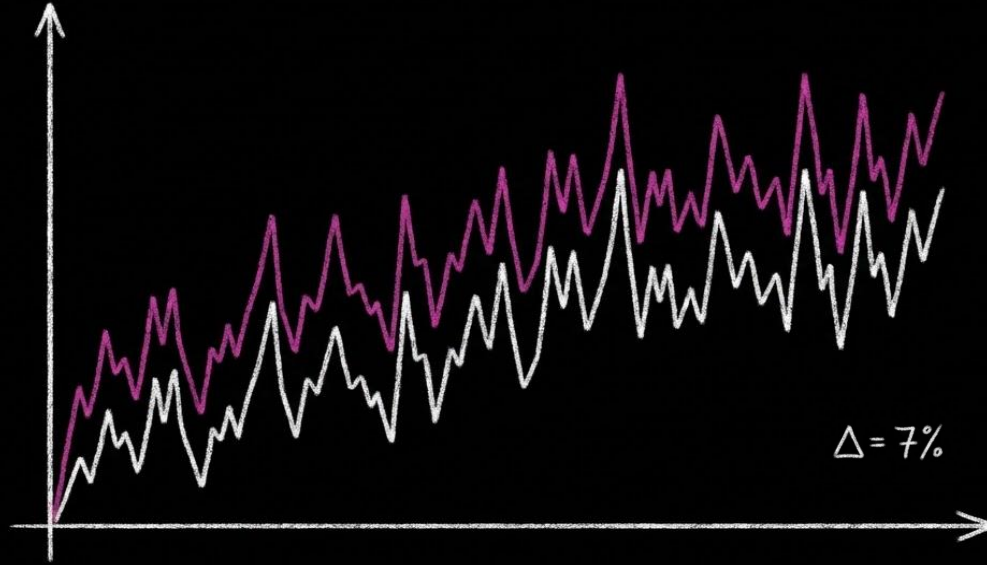
Metrics Shape Your Tooling

```
- id: metric.system.cpu.time
  attributes:
    - ref: cpu.mode
    - ref: cpu.logical_number
+   requirement_level: opt_in
```

```
cpu:  
  metrics:  
    system.cpu.time:  
      enabled: true  
      attributes: [cpu.mode, cpu.logical_number]
```

Lesson 4

Tooling Shapes **User Experience**



total - free - cached - buffers

total - available

[#42223](#)

Embrace breaking changes

name: system_otel

title: "System OpenTelemetry
Assets"

version: 0.2.5

conditions:

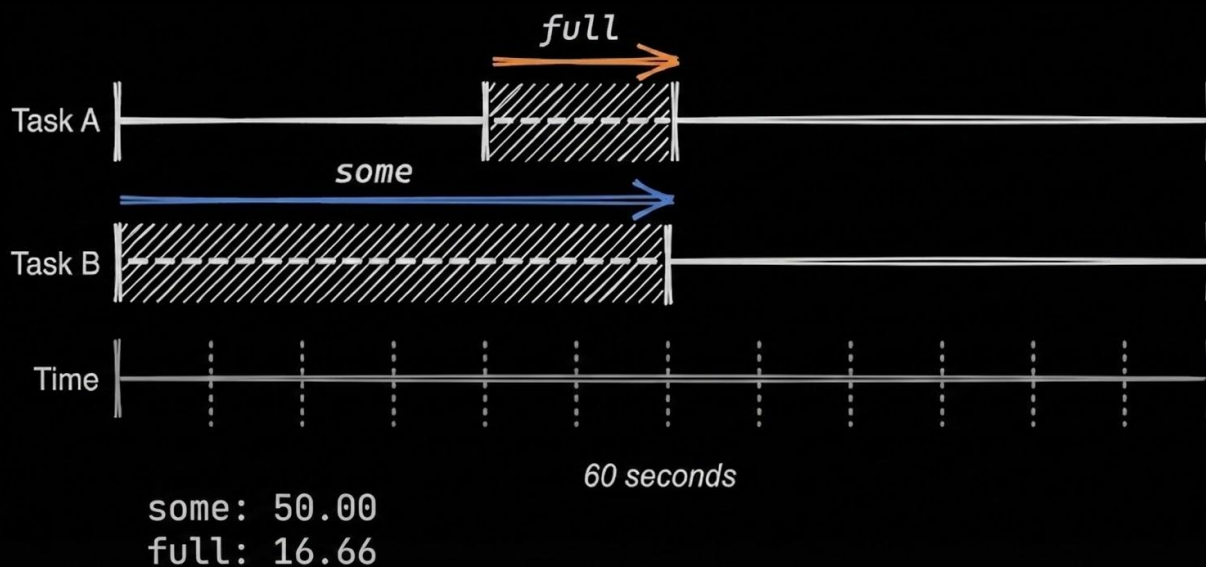
kibana:

version: "^8.18.0 || ^9.0.0"

Lesson 5

User Experience Drives Contributions

`system.{cpu,memory,io}.linux.pressure.*`



[#3068](#)

Thank you!

Huge kudos to @braydonk, @ChrsMark, @dmitryax, @frzifus
and @mx-psi